

Published in IET Generation, Transmission & Distribution
 Received on 10th October 2012
 Accepted on 10th March 2013
 doi: 10.1049/iet-gtd.2012.0589



Reconfigurable instrument for neural-network-based power-quality monitoring in 3-phase power systems

Martin Valtierra-Rodriguez¹, Rene de J. Romero-Troncoso^{1,2}, Arturo Garcia-Perez², David Granados-Lieberman¹, Roque A. Osornio-Rios¹

¹HSPdigital – CA Mecatronica, Facultad de Ingenieria Campus San Juan del Rio, Universidad Autonoma de Queretaro, Rio Moctezuma 249, 76807, San Juan del Rio, Queretaro, Mexico

²HSPdigital – CA Telematica/Procesamiento Digital de Señales, Division de Ingenierias, Campus Irapuato-Salamanca, Universidad de Guanajuato, Salamanca, Guanajuato, Mexico
 E-mail: raosornio@hspdigital.org

Abstract: From voltage and current signals it is possible to obtain relevant information for solving some problems in several industrial and scientific applications as power quality (PQ) monitoring, monitoring and diagnosis of electrical machines, electric systems protection and control. At present, the PQ monitoring, measure through a set of PQ indices (PQI), is an important topic for the industrial sector since a poor PQ, characterised by the presence of harmonics in the power line, produces irregular or wrong operation of protection systems, excessive neutral currents in 3-phase four-wire systems, overheating of motors, transformers, capacitor banks and wiring in general. The PQI calculation is performed by many techniques proposed in the literature; however, they do not have either good performance for transient signals or the requirements for satisfying the power standards. This work proposes the assessment of the PQI-based in neural networks for transient or stationary signals in 3-phase power systems without losing the power standard requirements. Besides, this work contributes to the industrial application field by allowing the continuous and online monitoring of the PQI thanks to the field programmable gate array implementation of the proposed methodology.

1 Introduction

Voltage and current signals are the major analysed signals on any electrical system. From these signals, it is possible to obtain relevant information for solving some problems in several industrial and scientific applications as: power quality (PQ) monitoring, monitoring and diagnosis of electrical machines, electric systems protection and control, among others [1].

Currently, the PQ monitoring is an important topic for the industrial sector since a poor PQ, characterised by the presence of harmonics in the power line [2], produces irregular or wrong operation of protection systems, excessive neutral currents in 3-phase four-wire systems, overheating of motors, transformers, capacitor banks and wiring in general, decreasing the useful life of electrical devices and so on. The measurement or assessment of the PQ is done through a set of PQ indices (PQI), which are the basis of PQ standards for illustrating the negative impact of electrical disturbances [3–6]. Besides, some PQI are used as parameters of diagnosis in monitoring and control tasks [7–9], where a correct and

opportune estimation of their values is crucial for these methodologies.

Basically, until now there have been identified two solutions for PQ monitoring, those reported in literature and commercially available products. In the literature, PQI are mostly obtained from the signal frequency spectrum and Parseval's theorem [10]; therefore the contributions of current and voltage harmonics are considered for the PQI calculation. Some techniques used for harmonic estimation are the fast Fourier transform (FFT), the Goertzel algorithm, the Chirp Z transform, the Welch algorithm, the Zoom FFT, amongst others [10]. Nevertheless, their disadvantages for assessing transient signals are well known. To face this issue, multiple studies propose new PQI for transient signals using time-frequency algorithms, such as bank filters, wavelet transforms, Kalman filters, short-time Fourier transform (STFT), parametric methods etc. [11–14]. However, for harmonic analysis and mainly for the IEC61000 [3–4] and EN 50160 [6] standard requirements, the aforementioned techniques present some difficulties; for instance, discrete wavelet analysis needs many decomposition levels to be able to measure all the

harmonics and interharmonics required by the standard; the bank filters have the same problem, one filter is needed for each harmonic or interharmonic; besides, additional processing is required to quantify the harmonic magnitude and phase. Although the STFT is the most useful solution for harmonic evaluation, it requires processing a time window of the sampled signal, where the signal is considered stationary, which is not true in transient phenomena.

Recently, the convenience of utilising neural networks (NN) to estimate the harmonic magnitudes and their time evolution has been recognised by many authors [15–20]. However, these works do not satisfy the standard requirements since they do not calculate interharmonics, and the harmonics calculated are fewer than the required by the standards; therefore the contributions of current and voltage harmonics are not well estimated and the PQI calculation is not possible. Besides, these works are implemented with specialised software in personal computers (PC), limiting their portability and computation speed, as an immediate consequence the online operation is not always possible.

On the other hand, commercially available equipments present some disadvantages in their performance; for instance, they estimate few electric parameters or have a specific set of functions to fulfil the necessities of specific applications, and the integration of tasks, different from those they are designed for, is limited; besides, in most of these equipments it is impossible to change characteristics such as sampling rate or number of acquired data, which is desirable in many applications. Field programmable gate array (FPGA) devices are the most promising technology to overcome these disadvantages thanks to their flexibility, reconfigurability, parallelism and velocity, as it has been demonstrated in applications of PQ monitoring, motor-condition diagnosis, machine-tool control and in many industrial applications [21].

This work proposes an NN-based methodology for the assessment of the PQI in 3-phase power systems. Besides, this methodology offers the capability of analysing transient or stationary signals without losing the power standard requirements unlike other methodologies. Finally, this methodology contributes to the industrial application field by allowing the continuous and online monitoring of the PQI thanks to its FPGA implementation. The validation and testing of the proposed methodology are carried out through simulation with a synthetic signal and with two real experiments in an electric motor, respectively.

2 Theoretical background

PQI are mostly obtained from the signal frequency spectrum and Parseval's theorem; therefore for the PQI calculation it is necessary the rightful harmonic estimation for current and voltage signals. The appropriate monitoring of harmonics is supervised by the IEC61000-4-7 standard [3], where the analysis is referred to signals described in (1); where the harmonic and interharmonic analysis is mostly done using the STFT.

$$y(t) = \sum_{m=0}^{\infty} \left[a_m \sin\left(\frac{m}{N} \omega t\right) + b_m \cos\left(\frac{m}{N} \omega t\right) \right] \quad (1)$$

where y is the signal amplitude in the time domain t ; ω is

the fundamental frequency $\omega = 2\pi f$; m is the number of the spectral line with respect to the fundamental component; N is the number of fundamental periods within the window width; a_m and b_m are the amplitude coefficients of the sine and cosine components in the Fourier series, respectively.

An alternative expression of (1) is

$$y(t) = \sum_{m=0}^{\infty} c_m \sin\left(\frac{m}{N} \omega t + \varphi_m\right) \quad (2)$$

with

$$c_m = \sqrt{a_m^2 + b_m^2}, \quad \phi_m = \arctan\left(\frac{a_m}{b_m}\right) \quad \text{if } b_m \geq 0, \quad (3)$$

$$\text{and } \phi_m = \pi + \arctan\left(\frac{a_m}{b_m}\right) \quad \text{if } b_m < 0$$

where φ_m is the phase angle and c_m is the amplitude of the component with frequency f_m

$$f_m = \frac{m}{N} f \quad (4)$$

This standard defines the size of the signal that has to be analysed in order to deliver a result, for both 50 and 60 Hz systems; this is 10 and 12 cycles of the signal or basically a time window of 200 ms. This window size allows having a resolution of 5 Hz for both supply systems. According to the standard all the harmonics until the 40th have to be measured; therefore the range of analysis is from 0 to 2000 and 2400 Hz, respectively. Equation (1) is modified with respect to each supply system to accomplish the range of analysis and the specified resolution. When the harmonic values are correctly obtained it is possible the PQI calculation in the frequency domain. The PQI calculated in this work for voltage and current signals in 1-phase and 3-phase systems are shown in Table 1.

3 NN-based PQI

The harmonic estimation is done by a single layer NN called adaptive linear neuron (ADALINE) [22]. For achieving a resolution of 5 Hz and up to the 40th harmonic the ADALINE is arranged as shown in Fig. 1. The only input to ADALINE is the fundamental frequency $\omega = 2\pi f$ for the time t , this is multiplied by a numerical factor m/N that represents the harmonic or interharmonic order, and it is evaluated by a sine or cosine function. Then, the outputs of the previous step are pondered by the weights a_m and b_m that represent an estimated value of the amplitude for the sine or cosine components in a Fourier series, respectively. The variable y_k is the sum of the previous outputs, which represents an estimate of y . A learning algorithm is given to ADALINE for processing the error $(y - y_k)$ and thus to modify the weights in order to force the error to zero. It has to be noted that the error is equal to zero when all the harmonics of y are modelled by ADALINE; therefore the weights are an exact measure of the harmonic magnitudes of y . The output of ADALINE y_k

Table 1 Power quality indices

1-Phase PQI	Equations	3-phase PQI	Equations
root mean square (RMS) [23]	$I_{rms} = \frac{1}{\sqrt{2}} \sqrt{\sum_{n=1}^N I_n^2} \quad (5)$	equivalent I and V [24]	$I_e = \sqrt{\frac{I_a^2 + I_b^2 + I_c^2}{3}} \quad (11)$
	$V_{rms} = \frac{1}{\sqrt{2}} \sqrt{\sum_{n=1}^N V_n^2}$		$V_e = \sqrt{\frac{V_{ab}^2 + V_{bc}^2 + V_{ca}^2}{9}}$
total harmonic distortion (THD) [23]	$THD_{fund} = \frac{\sqrt{\sum_{n=2}^N I_n^2}}{I_1} \times 100\% \quad (6)$	equivalent THD [24]	$THD_{eI} = \frac{I_{eH}}{I_{e1}}$
	$THD_{fund} = \frac{\sqrt{\sum_{n=2}^N V_n^2}}{V_1} \times 100\%$		$I_{eH} = \sqrt{\frac{I_{aH}^2 + I_{bH}^2 + I_{cH}^2}{3}}$
			$I_{e1} = \sqrt{\frac{I_{a1}^2 + I_{b1}^2 + I_{c1}^2}{3}} \quad (12)$
			$THD_{eV} = \frac{V_{eH}}{V_{e1}}$
			$V_{eH} = \sqrt{\frac{V_{abH}^2 + V_{bcH}^2 + V_{caH}^2}{9}}$
			$V_{e1} = \sqrt{\frac{V_{ab1}^2 + V_{bc1}^2 + V_{ca1}^2}{9}}$
apparent power (S) [23]	$S = V_{rms} I_{rms} \quad (7)$	symmetrical components and unbalance [1]	$\begin{bmatrix} U_0 \\ U_{pos} \\ U_{neg} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} U_{ab} \\ U_{bc} \\ U_{ca} \end{bmatrix}$
real power (P) [23]	$P = \frac{1}{2} \sum_{n=1}^N V_n I_n \cos(\theta_n - \varphi_n) \quad (8)$		where $a = \exp[j(2\pi/3)]$ and
reactive power (Q) [23]	$Q = \frac{1}{2} \sum_{n=1}^N V_n I_n \sin(\theta_n - \varphi_n) \quad (9)$		$\begin{aligned} U_{ab} &= U_b - U_a \\ U_{bc} &= U_c - U_b \\ U_{ca} &= U_a - U_c \end{aligned}$
power factor (PF) [23]	$PF = \frac{P}{S} \quad (10)$	power factor 3-phase [24]	$\% \text{UMB} = \frac{ U_{neg} }{ U_{pos} } \times 100\% \quad (13)$
			$PF_{3\theta} = \frac{P_T}{S_e}$
			$P_T = P_a + P_b + P_c$
			$S_e = 3V_e I_e \quad (14)$

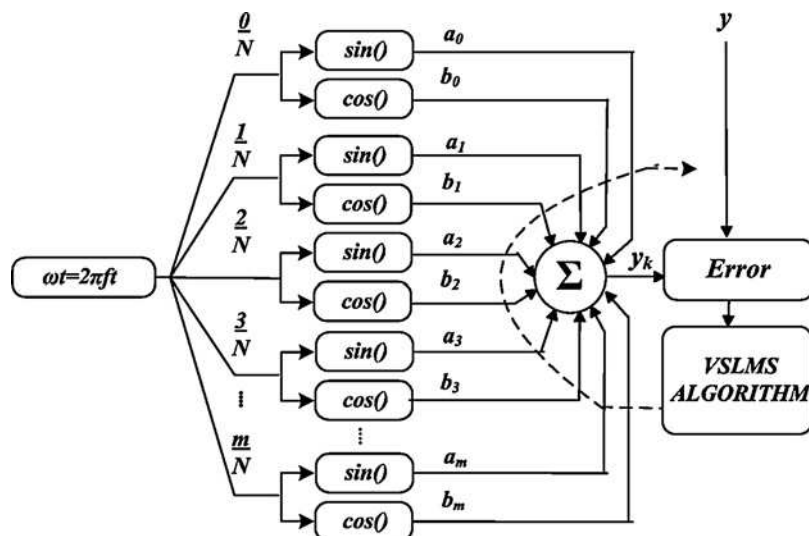


Fig. 1 Neural network for analysing transient and stationary signals according to the standard requirements

is given in (15)

$$y_k(t) = w^T x \quad (15)$$

with

$$w(k) = \begin{bmatrix} a_0 \\ b_0 \\ a_1 \\ b_1 \\ \vdots \\ a_m \\ b_m \end{bmatrix} \quad x(k) = \begin{bmatrix} \sin\left(\frac{0}{N} 2\pi ft\right) \\ \cos\left(\frac{0}{N} 2\pi ft\right) \\ \sin\left(\frac{1}{N} 2\pi ft\right) \\ \cos\left(\frac{1}{N} 2\pi ft\right) \\ \vdots \\ \sin\left(\frac{m}{N} 2\pi ft\right) \\ \cos\left(\frac{m}{N} 2\pi ft\right) \end{bmatrix} \quad (16)$$

The learning algorithm performs an automatic small adjustment for each input–output combination in the values of the weights until that it obtains the correct outputs. The used algorithm to minimise the error is the variable step-size least-mean-square (VSLMS) algorithm [25]; this can be described as follows

$$e(k) = d(k) - w^T(k)x(k) = y(k) + v(k) - w^T(k)x(k) \quad (17)$$

$$w(k+1) = w(k) + \mu(k)e(k)x(k) \quad (18)$$

where $e(k)$ is the instantaneous error, $x(k) = [x(k), \dots, x(k-N+1)]^T$ is the input signal vector, N is the tap-length of the filter,

$(\cdot)^T$ denotes the vector transpose operator, $w(k) = [w_0(k), \dots, w_{N-1}(k)]^T$ is the coefficient or weight vector, $d(k)$ is the desired signal, $y(k)$ is the output of the unknown system and $v(k)$ is the system noise. The weight update recursion is given by (25) and the updates step-size (μ) expression is

$$\mu(k+1) = \alpha\mu(k) + \gamma_s \left\| \hat{P}(k) \right\|^2 e^2(k) \quad (19)$$

where $0 < \alpha < 1$ and $\gamma_s > 0$ are the learning rates. $\hat{P}(k)$ is a smoother gradient vector by weighted-averaging as

$$\hat{P}(k) = \beta \hat{P}(k-1) + (1-\beta)e(k)x(k) \quad (20)$$

where β is a smoothing factor that is close to unity and in the range $0 < \beta < 1$. $\|\cdot\|^2$ denotes the squared Euclidian norm operation. With the values a_m and b_m calculated by ADALINE, the magnitudes and phases are obtained by (3). The adjustment of the weights is for each new sample of the signal acquired; therefore the evolution of the harmonics that compose the signal through time is known at every sample of the signal. On the other hand, the harmonic values are used to estimate the PQI through time as well, since they depend directly of the harmonic content; therefore the values of PQI are calculated for each new sample of the acquired signal. The mathematical formulae are the same that shown in Table 1; however, the results are by sample not by time window, with that, their time resolution is improved for transient and stationary signals.

4 3-Phase monitoring system

The proposed methodology and the features of the developed instrument satisfy the standard requirements. The system has two main blocks: (1) instrumentation system and (2) FPGA-based signal processing unit, as shown in Fig. 2.

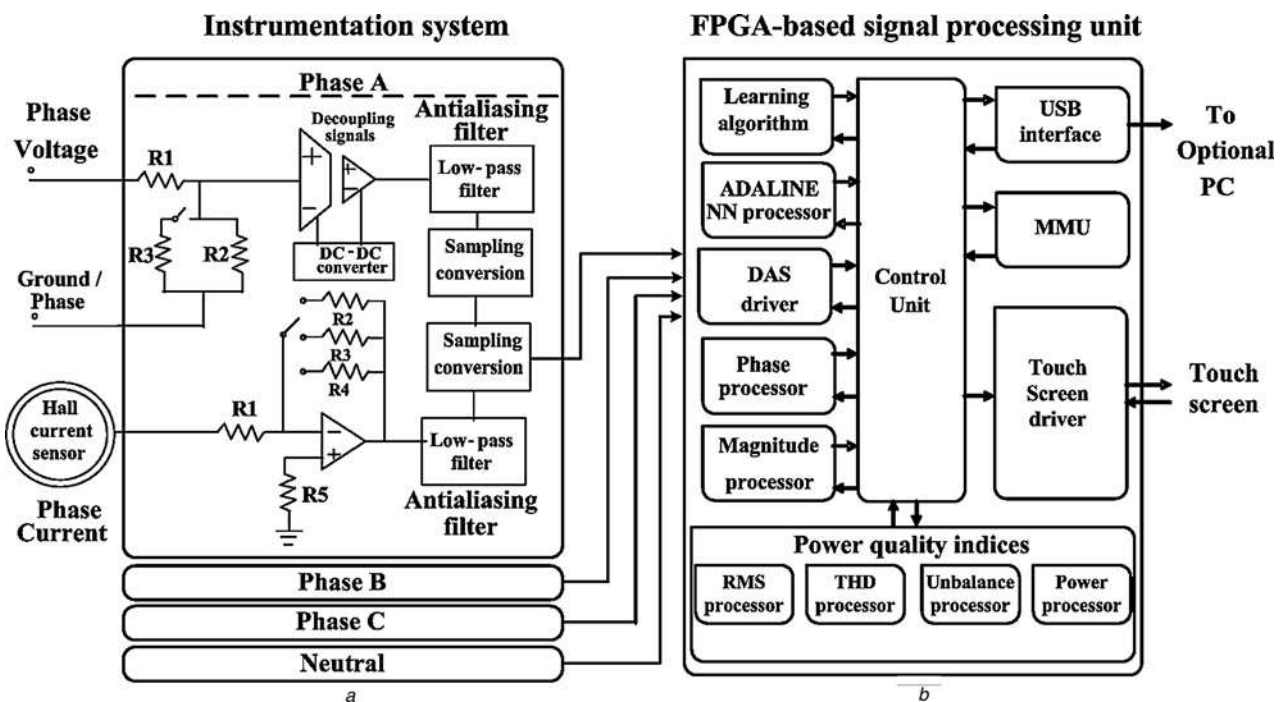


Fig. 2 Overall system

a Instrumentation for 3-phase monitoring in a power system
b Block diagram of the FPGA-based signal processing unit

4.1 Instrumentation system

The block diagram of the developed system for sensing voltage and current signals in one phase of a 3-phase power supply connection is shown in Fig. 2a. For monitoring the current signal on each phase and ground, four hall-effect sensors model L08P050D15, from Tamura Corporation. On the other hand, the voltage signal is monitored by a voltage divider and a precision isolation amplifier model ISO124PND in order to obtain galvanic isolation between the power system and the proposed monitoring system. The proposed system features allow measuring voltage between any two phases or one phase and ground. The selectable gain can be changed in order to measure up to 200% of the nominal voltage. The data-acquisition system (DAS) for the current and voltage signals has a sampling stage consisting in a 16-bit 4-channel serial-output sampling analog-to-digital converter ADS8341 with a sampling rate of 6 kHz. The digital information is processed by the algorithms implemented in an FPGA device.

4.2 FPGA-based signal processing unit

The FPGA-based implementation of the signal processing unit is shown in Fig. 2b. The proposed methodology is completely implemented in an FPGA. To keep the overall cost of the system low and to comply with the processing speed requirements, all the processing cores (ADALINE NN, VSLMS, USB interface, PQI etc.) were completely developed by the authors, and no commercially processing cores are used. USB interface allows communicating the instrument with an optional PC for storage and additional processing. The DAS driver interfaces the processing unit with the instrumentation system to acquire the 3-phase current and voltage signals from the instrumentation system block. The modules for the hardware implementation of the proposed methodology are as follows: the ADALINE NN processor performs (15), the learning algorithm VSLMS processor computes (17) and the magnitude and phase processor calculates (3), respectively; while RMS, THD, unbalance and power processors perform (5)–(14), respectively. All computations are executed each sampling period with a sampling frequency of 6000 Hz, meaning that all the PQI computations have to be performed in less than $1/6000 \text{ Hz} \approx 166.6 \mu\text{s}$ to keep the real-time operation. To accomplish the time requirements, these blocks are implemented with optimised digital structures and the task is performed in about $104.85 \mu\text{s}$ which is approximately 1.5 times faster than the required time. The optimisation is also because of the use of the coordinate rotation digital computer equations for the obtaining of sine/cosine and magnitude/phase [26]. The digital touch panel driver generates the plots for all the different PQI calculated for voltage or current signals, and sends them to the RBD-LTM touch screen [27] for display purposes only; in addition, this driver allows select the PQI to be shown in the touch panel thanks to the programmed touch commands. The memory management unit synchronises input and output data for all the internal processes. The board is a proprietary Spartan 3E XC3S1600E FPGA [28] with a working clock frequency of 48 MHz; however, the system has a maximum frequency operation of 87.5 MHz. The resource usage synthesis report and the space distribution for the selected FPGA are shown in Table 2. It is appreciated that the NN-based PQI processors consume the

Table 2 Synthesis report for resources usage and space distribution into the FPGA

Resources	Used	Available	%	
			module	%
slices	23,308	29,504	DAS and USB driver	10
			NN-based PQI processors	52
			digital touch panel driver	17
			79	
block RAM	522 K	648 K	80.5	
bits				
18 × 18b mult	8	36	22	

most of resources since they perform the calculus for all the PQI in 3-phase power systems for current and voltage signals.

5 Experimentation and results

This section describes the experiments carried out to validate and to test the functionality of the proposed methodology for transient and stationary signals; in addition, its operation is compared with the 3-phase PQ analyser fluke 435 [29] by using the fluke current clamps i200s [30] for comparison and calibration purposes.

5.1 Validation

The methodology has been tested with synthetic signals in order to have a priori knowledge of the true values of the harmonics and interharmonics through time. Therefore the difference or error between the true value and the obtained value can be measured. The PQI are not analysed in this case since the PQI values are directly related with the harmonic content. The generated synthetic signal is the sum of harmonics and interharmonics with different amplitudes, which are added to different times in concordance with (21)

$$y_s(t) = \sum_{k=0}^K [A_k \sin(2\pi F_{hk}t) + A_k \sin(2\pi F_{ik}t)] u(t - k\Delta t) \quad (21)$$

where $u(t - k\Delta t)$ is the step function with a displacement $\Delta t = 80 \text{ ms}$ for $k = 0, 1, 2, 3, 4 \dots K = 12$ i $t = 1 \text{ s}$.

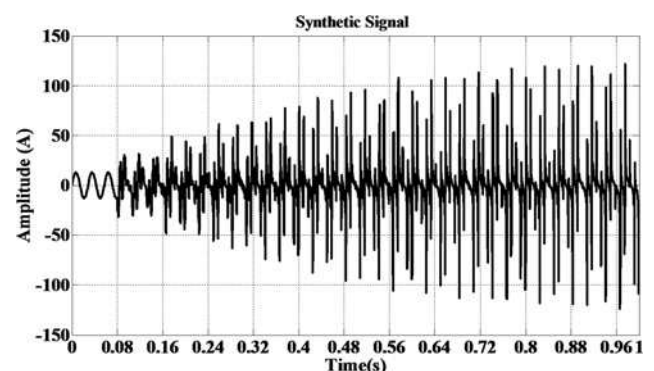


Fig. 3 Synthetic signal

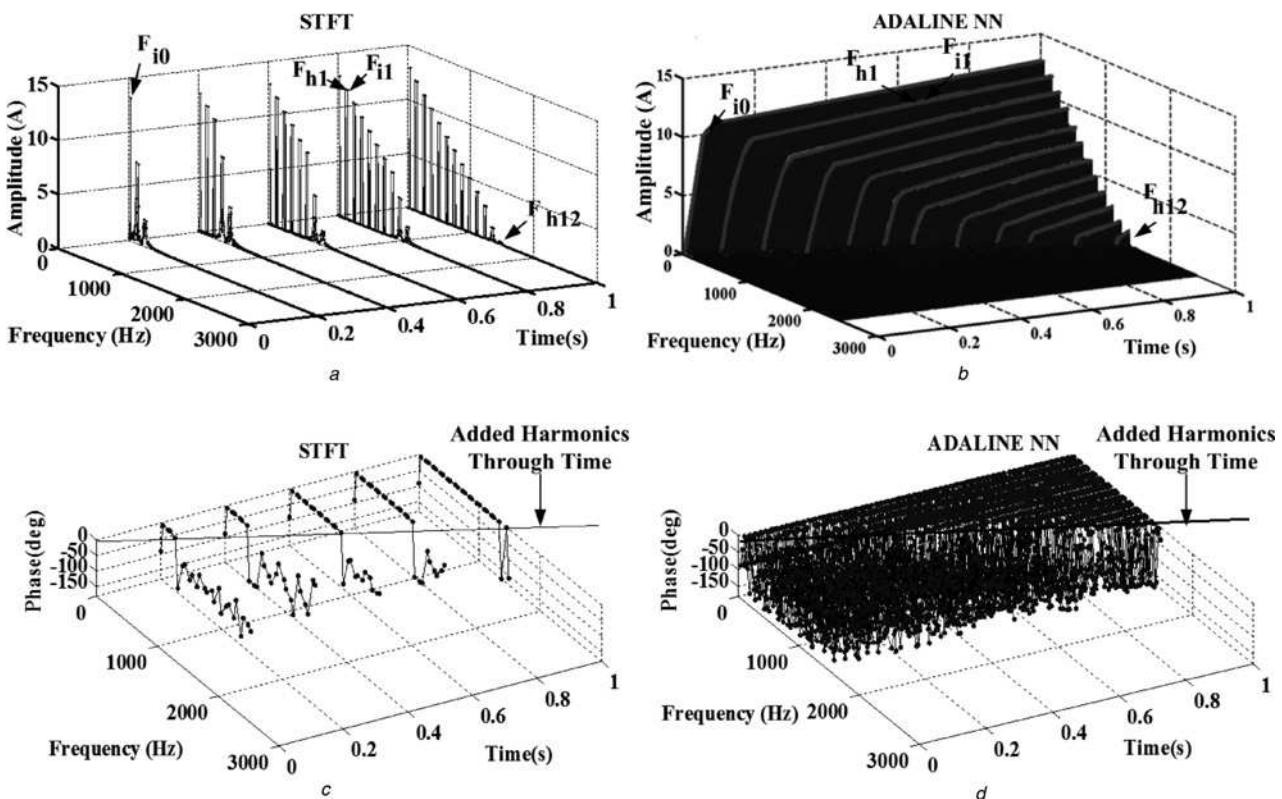


Fig. 4 Analysis of the synthetic signal

- a STFT: amplitude results
- b ADALINE NN: amplitude results
- c STFT: phase results
- d ADALINE NN: phase results

The amplitudes of harmonics $F_{hk} = 120k$ and interharmonics $F_{ik} = 120k + 35$ are $A_k = 13k$, respectively; the phase for all of them is zero. The synthetic signal is shown in Fig. 3.

Figs. 4a and b show the STFT and ADALINE NN results for the magnitudes, and Figs. 4c and d show the phase results, respectively. The average error in amplitude of the harmonic estimation techniques is 27.20% for STFT and 1.89% for ADALINE NN, and for phases the average errors are 26.17% and 1.13%, respectively.

5.2 Experiment setup

The developed instrument can be seen in Fig. 5a and the experimental setup is shown in Fig. 5b. This instrument and the Fluke 435 for comparison purposes are connected for monitoring an electric motor during the startup transient and the steady state when it is started up in two different ways, being the first one a full-voltage starter and the second one a variable speed drive. In the two cases of study, only few PQI are shown in plots for reasons of

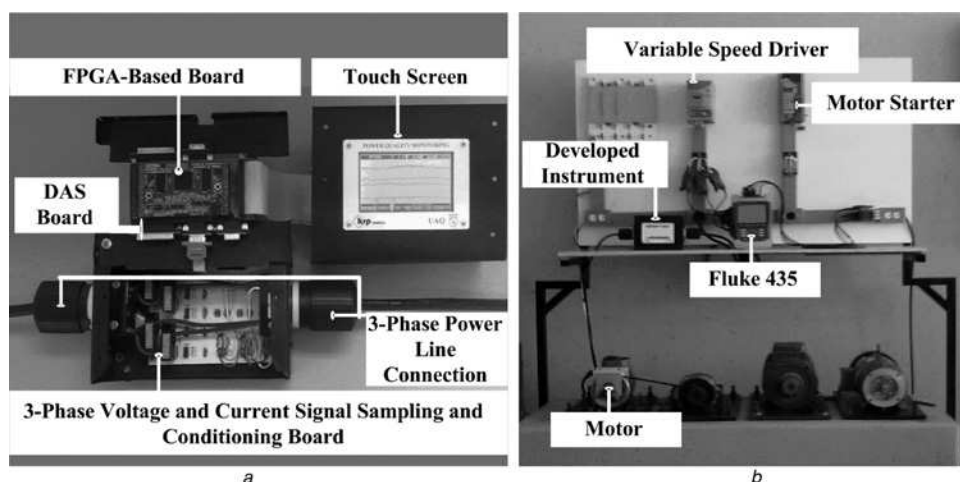


Fig. 5 Different study cases

- a Developed instrument
- b Experiment setup

Table 3 PQI results for the study cases

Study cases	I				II			
	NN-based PQI		Fluke 435		NN-based PQI		Fluke 435	
	ST	SS	ST	SS	ST	SS	ST	ST
Vrms <i>A, B</i> and <i>C</i> phase	125.2	125.6	125.6	125.8	105.1	144.3	108.2	145.3
	126	126.4	125.9	126.2	102.4	143.8	107.5	143.6
	124.7	125.4	125.0	125.3	105.8	142.8	108.5	144.6
Irms <i>A, B</i> and <i>C</i> phase	7.8	2.05	7.5	2.04	1.80	2.04	1.88	2.02
	7.62	2.01	7.33	1.99	1.81	2.05	1.88	2.03
	7.58	1.93	7.30	1.91	1.83	2.04	1.89	2.02
equivalent V	107.2	124.8	NA	NA	103.2	143.5	NA	NA
equivalent I	7.75	2.17	NA	NA	1.81	2.01	NA	NA
THDV <i>A, B</i> and <i>C</i> phase	26.2	1.92	24.7	1.88	75.2	10.7	71.2	10.9
	27.7	1.51	26.12	1.55	71.3	10.8	69.8	10.8
	26.8	2.50	25.6	2.54	74.3	10.35	70.5	10.4
THDI <i>A, B</i> and <i>C</i> phase	40.5	5.3	38.6	5.3	43.5	3.3	40.7	3.3
	30.2	6.2	25.2	6.1	44.1	3.1	41.3	3.2
	26.2	7.9	23.8	7.96	49.8	5.15	48.3	5.1
equivalent THDV	27.1	2.12	NA	NA	73.2	10.5	NA	NA
equivalent THDI	32.8	6.6	NA	NA	46.4	4.1	NA	NA
U_0 I	0.035	0.02	0.0	0.0	0.7	0.3	8.0	7.0
U_{pos} I	39.8	35.71	40.0	11.0	0.36	0.32	6.0	3.0
U_{neg} I	0.8	0.7	1.0	1.0	0.8	0.7	8.0	7.0
%UMB I	2.01	1.96	2.5	9.09	222.2	218.75	133.3	233.3
U_0 V	2.81	2.67	2.61	2.68	5.23	3.18	5.19	3.16
U_{pos} V	304.8	441.6	447.6	447.6	324.15	439.8	447.6	447.6
U_{neg} V	3.37	3.65	3.29	3.63	4.1	4.06	3.9	4.03
%UMB V	1.1	0.83	0.74	0.81	1.26	0.92	0.87	0.9
S <i>A, B</i> and <i>C</i> phase	976.5	257.48	942	256.6	189.18	294.3	203.4	293.5
	960.1	254.1	922.8	251.1	185.3	294.79	202.1	291.5
	945.2	242.1	912.5	239.3	193.6	291.3	205.1	292.1
P <i>A, B</i> and <i>C</i> phase	−620	−215	−634	−212	−120	−217	−143	−215
	−612	−211	−628	−214	−125	197	140	194
	−598	−198	−605	−194	−123	203	145	200
Q <i>A, B</i> and <i>C</i> phase	−454	−143	−432	−140	−133	−126	−128	−122
	−461	−121	−444	−118	114	122	108	120
	−497	−132	−482	−129	119	139	115	138
FP <i>A, B</i> and <i>C</i> phase	−0.63	−0.83	−0.67	−0.82	−0.63	−0.73	−0.70	−0.73
	−0.63	−0.83	−0.68	−0.85	−0.67	0.66	0.69	0.66
	−0.63	−0.81	−0.66	−0.81	−0.63	0.69	0.70	0.68
FP 3-phase	−0.63	−0.82	NA	NA	−0.65	0.65	NA	NA

NA: not available.

ST: startup transient.

SS: steady state.

space; however, all of them are available in Table 3 since they are obtained in a parallel way.

5.3 Study case I: full-voltage starter

In this case, a motor of 1-hp (746 W) is started by a full-voltage starter. The test embraces a transient and a stationary signal during the startup transient and steady state of the motor, respectively. Figs. 6*a* and *b* show the current and voltage signals, respectively; they also show an expanded part of the signal for display purposes only. Fig. 7 shows the results for the *P*, *Q* and *S* indices.

5.4 Study case II: variable speed driver

The 1-hp motor is also used, fed by the variable speed driver with a speed profile from 0 to 60 Hz in 5 s. Figs. 6*c* and *d* show the current and voltage signals, respectively. Fig. 8 shows the results for the THDV indices.

5.5 Analysis and discussion

The validation stage clearly shows how the STFT analysis gives a result at the end of the analysed window in Fig. 4*a*, where in addition, the results are given and considered stationary and any other information about the signal through time is proportionate. On the other hand, the proposed methodology shows where the harmonic or interharmonic appears, as well as their time evolution as shown in Fig. 4*b*. For display purposes only, Figs. 4*c* and *d* show just the phase results for the existent harmonics into the synthetic signal. The black line on the figures is placed to indicate the apparition of each harmonic or interharmonic into the synthetic signal through time; besides, it shows that when the harmonic has appeared the phase is correctly estimated; if there is no harmonic content, the values of the phases are oscillating and only when the amplitude of the harmonic is significant or non-zero, the value of the phase has to be considered.

On the other hand, for showing the effectiveness of the developed instrument, the two most common ways for starting a motor are analysed. For the startup transient in both study cases, the advantages of ADALINE NN are

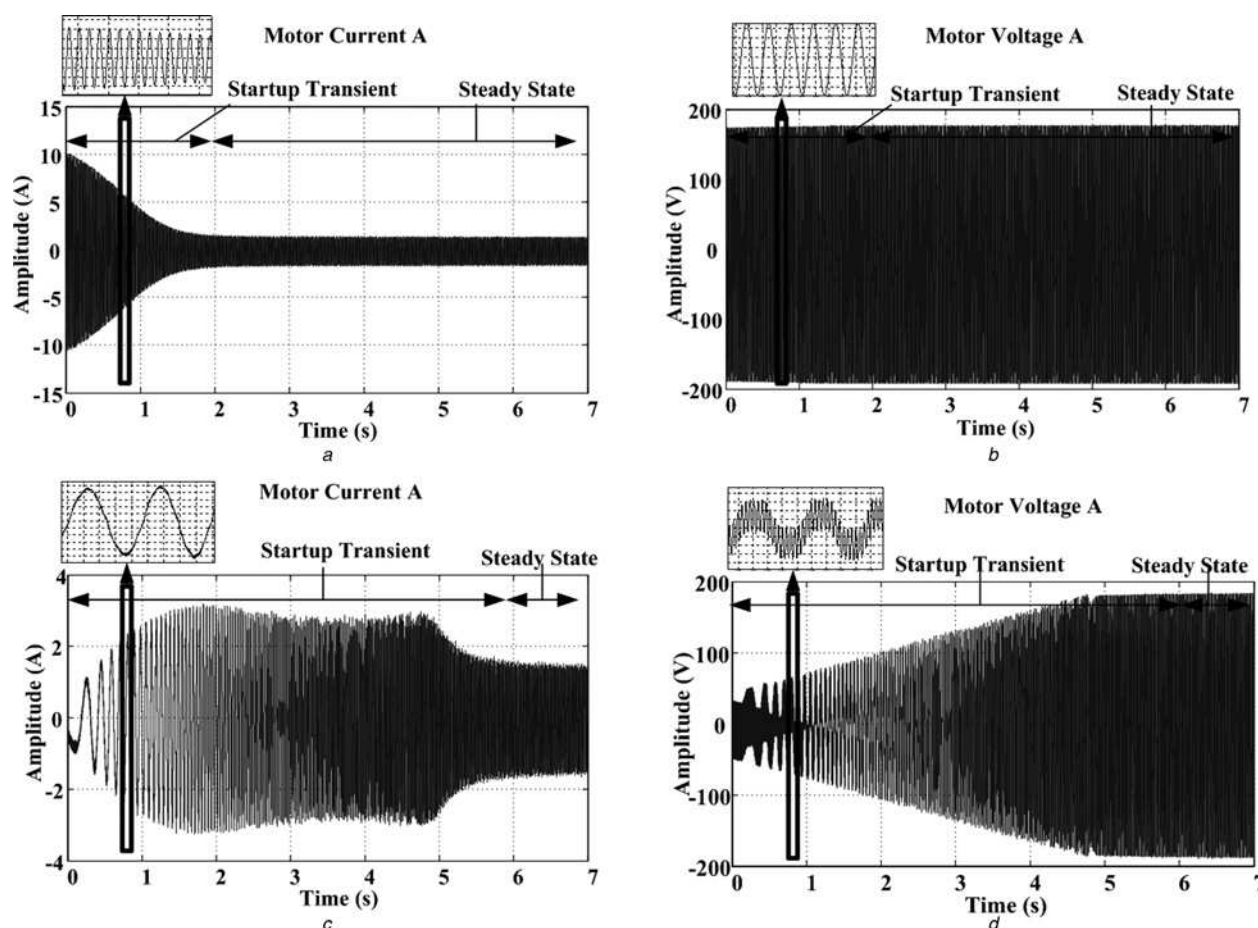


Fig. 6 Study cases analysed with the proposed instrument and the Fluke 435

- a Full voltage starter: current signal
- b Full-voltage starter: voltage signal
- c Variable speed driver: current signal
- d Variable speed driver: voltage signal

clearly appreciated in Figs. 7 and 8 since they show a more detailed evolution of the PQI through time. It is important to emphasise that STFT gives five spectra per second according to the standard, whereas the ADALINE NN gives 6000 spectra per second, one at each sample; therefore 6000 values of each PQI are given per second. For inherent reasons, STFT is not able to improve its time resolution

without affecting the resolution in frequency or contrariwise, for example, for improving twice the time resolution the size of the window has to be reduced to 100 ms; however, with this window size, the frequency resolution will be of 10 Hz which does not satisfy the standard requirements. Therefore STFT is not able to reach the time resolution of ADALINE NN. When the signals are

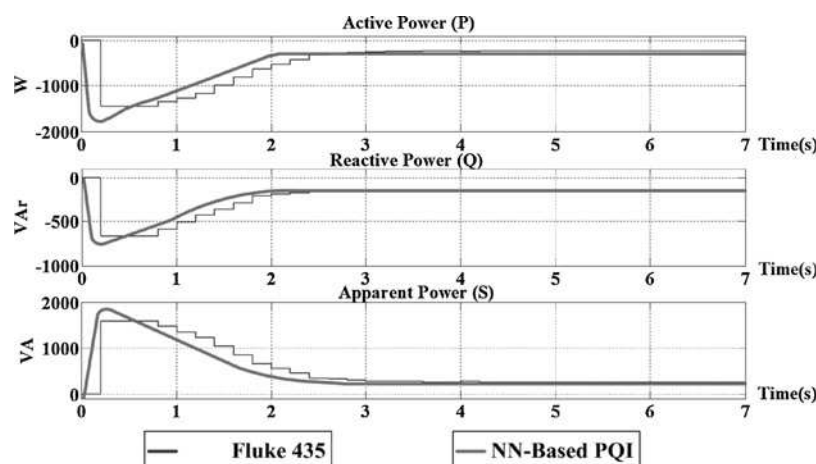


Fig. 7 Results for P , Q and S powers

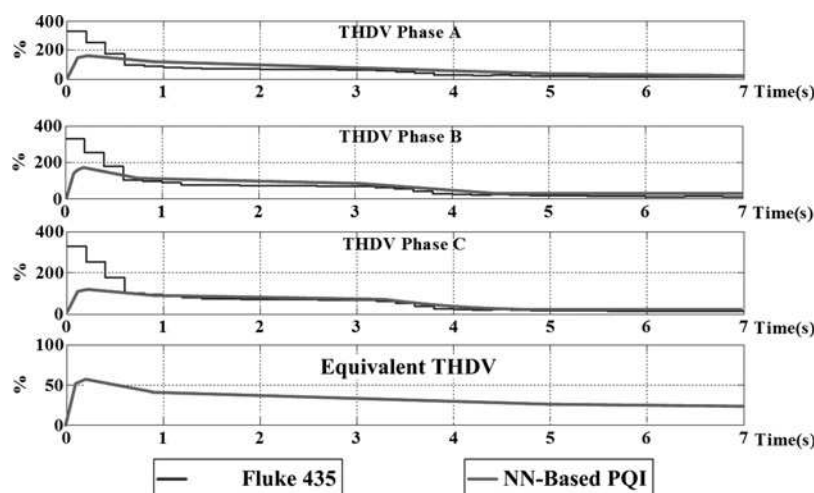


Fig. 8 Results for 1-phase and 3-phase THD in voltage signals

in steady state, both instruments for the PQI calculation do not present a notable difference since the STFT gives good results for stationary signals; however, the ADALINE NN time resolution is better.

Table 3 shows the PQI results for both study cases, which contains the results of the proposed instrument and the commercial equipment for transient and steady state in concordance with Fig. 6. It is convenient to mention that the values are just the average of the results of each instrument since the proposed instrument and the commercial equipment give 6000 and 5 values per second, respectively. In steady state, both instruments show similar results with no more difference than 2.5%; however, in transient state the results of the proposed instrument are more reliable than the other ones as it was shown in the validation stage.

The analysis and monitoring of transient signals are the aim of the proposed instrument. When a transient occurs throughout the analysed time window, the commercial equipments by means of STFT give an average value of the harmonics and not their evolution as the proposed instrument.

6 Conclusions

This paper presents a novel ADALINE NN-based methodology for PQI estimation in 3-phase power systems, which gives the time-evolution of harmonics and interharmonics for amplitudes and phases complying with the power standard requirements, contrary to other reported works that do not satisfy them; therefore the PQI are correctly estimated and their time-evolution is also known with a great time resolution. Besides, this work improves the standard monitoring techniques of other methodologies or instruments by providing the user with a powerful tool for monitoring the PQI in transient or steady states. On the other hand, the contribution to the industrial application field of this methodology is that its implementation into the FPGA allows the continuous and online monitoring of the PQI. Furthermore, the FPGA reconfigurability allows add more PQI or have a set of functions to fulfil the necessities of different applications unlike commercially available equipment where the integration of tasks is not possible. Besides, the development of proprietary hardware signal-processing cores helps to keep the overall cost low. Finally, the developed instrument and the proposed

methodology can be utilised for further research development in the area of PQ monitoring by adding control and diagnosis tasks for each PQI through time.

7 References

- Romero-Troncoso, R.J., Cabal-Yepez, E., Garcia-Perez, A., Osornio-Rios, R.A., Alvarez-Salas, R., Granados-Lieberman, D.: 'Reconfigurable instrument for power quality monitoring in 3-phase power systems'. Proc. IEEE Int. SDEMPED, Bologna, Italy, September 2011, pp. 186–191
- Bollen, M.H.J.: 'What is power quality?', *Electr. Power Syst. Res.*, 2003, **66**, (1), pp. 5–14
- IEC 61000-4-7: 'Testing and measurement techniques—general guide on harmonics and interharmonics measurements and instrumentation, for power supply systems and equipment connected thereto'. 2002
- IEC 61000-4-30: 'Testing and measurement techniques power quality measurement methods'. 2003
- IEEE Std. 1159: 'IEEE recommended practices for monitoring electric power quality'. 1995
- European Standard EN 50160: 'Voltage characteristics of electricity supplied by public distribution systems'. 2002
- Zhang, G., Xu, W.: 'Estimating harmonic distortion levels for systems with random-varying distributed harmonic-producing loads', *IET Gener. Transm. Distrib.*, 2008, **2**, (4), pp. 847–855
- Sahni, M., Lee, W.J.: 'Classification of severity of low-voltage motor coil arcing fault using statistical techniques', *IET Gener. Transm. Distrib.*, 2009, **3**, (1), pp. 75–85
- Wang, F., Duarte, J.L., Hendrix, M.A.M.: 'Design and analysis of active power control strategies for distributed generation inverters under unbalanced grid faults', *IET Gener. Transm. Distrib.*, 2010, **4**, (8), pp. 905–916
- Granados-Lieberman, D., Romero-Troncoso, R.J., Osornio-Rios, R.A., Garcia-Perez, A., Cabal-Yepez, E.: 'Techniques and methodologies for power quality analysis and disturbances classification in power systems: a review', *IET Gener. Transm. Distrib.*, 2011, **5**, (4), pp. 519–529
- Gu, Y.H., Bollen, M.H.J.: 'Time-frequency and time-scale domain analysis', *IEEE Trans. Power Deliv.*, 2000, **15**, (4), pp. 1279–1284
- Duque, C.A., Silveira, P.M., Ribeiro, P.F.: 'Visualizing time-varying harmonics using filter banks', *Electr. Power Syst. Res.*, 2011, **81**, (4), pp. 974–983
- Yilmaz, A.S., Alkan, A., Asyali, M.H.: 'Applications of parametric spectral estimation methods on detection of power system harmonics', *Electr. Power Syst. Res.*, 2008, **78**, (4), pp. 683–693
- Garcia-Perez, A., Romero-Troncoso, R.J., Cabal-Yepez, E., Osornio-Rios, R.A.: 'The application of high-resolution spectral analysis for identifying multiple combined faults in induction motors', *IEEE Trans. Ind. Electron.*, 2011, **58**, (5), pp. 2002–2010
- Bertoluzzo, M., Buja, G.S., Castellan, S., Fiorentin, P.: 'Neural network technique for the joint time–frequency analysis of distorted signal', *IEEE Trans. Ind. Electron.*, 2003, **50**, (6), pp. 1109–1115

- 16 Sarkar, A., Choudhury, S.R., Sengupta, S.: 'A self-synchronized ADALINE network for on-line tracking of power system harmonics', *Measurement*, 2011, **44**, (4), pp. 784–790
- 17 Lin, H.C.: 'Intelligent neural network-based fast power system harmonic detection', *IEEE Trans. Ind. Electron.*, 2007, **54**, (1), pp. 43–52
- 18 Chang, G.W., Chen, C.I., Teng, Y.F.: 'Radial-basis-function-based neural network for harmonic detection', *IEEE Trans. Ind. Electron.*, 2010, **57**, (6), pp. 2171–2179
- 19 Temurtas, F., Gunturkun, R., Yumusaka, N., Temurtas, H.: 'Harmonic detection using feed forward and recurrent neural networks for active filters', *Electr. Power Syst. Res.*, 2004, **72**, (1), pp. 33–40
- 20 Temurtas, H., Temurtas, F.: 'An application of neural networks for harmonic coefficients and relative phase shifts detection', *Expert Syst. Appl.*, 2011, **38**, (4), pp. 3446–3450
- 21 Monmasson, E., Idkhajine, L., Cirstea, M.N., Bahri, I., Tisan, A., Naouar, M.W.: 'FPGAs in industrial control applications', *IEEE Trans. Ind. Inf.*, 2011, **7**, (2), pp. 224–243
- 22 Dash, P.K., Swain, D.P., Liew, A.C., Rahman, S.: 'An adaptive linear combiner for on-line tracking of power system harmonics', *IEEE Trans. Power Syst.*, 1996, **11**, (4), pp. 1730–1735
- 23 Lara-Cardoso, J., Romero-Troncoso, R.J.: 'Low-cost power harmonics analyzer of nonlinear loads based on FPGA'. Proc. IEEE Instrumentation and Measurement Technology, Vancouver, Canada, May 2008. pp. 730–735
- 24 IEEE Std. 1459-2000: 'Definitions for the measurement of electric quantities under sinusoidal, nonsinusoidal, balanced, or unbalanced conditions'. 2000
- 25 Jeng-Kuang, H., Yuan-Ping, L.: 'Variable step-size LMS algorithm with a gradient-based weighted average', *IEEE Signal Process. Lett.*, 2009, **16**, (12), pp. 1043–1046
- 26 Vachhani, L., Sridharan, K., Meher, P.K.: 'Efficient FPGA realization of CORDIC with application to robotic exploration', *IEEE Trans. Ind. Electron.*, 2009, **56**, (12), pp. 4915–4929
- 27 Touch LTM User Manual, Terasic Technologies Inc., 2009
- 28 Data Sheet Spartan 3E FPGA family, Xilinx Corporation, 2009
- 29 Fluke 434/435 Three Phase Power Quality Analyzer user's manual, Fluke Corporation, 2006–2008
- 30 Fluke Current Clamp i200s Instruction Sheet, Fluke Corporation, 1998–2005